

Phytosanitary irradiation is the treatment pathway unlocking trade opportunities across the globe. It reduces harmful impacts on the environment and protects Australia's biosecurity. To learn more, please watch the videos and download the factsheets.

Food irradiation for safe trade

Title: Food irradiation for safe trade

Access to local and imported fruit and vegetables supports our dietary needs, agriculture industries and economies. Produce may contain pests that can harm environments and agriculture industries.

For trade to occur, produce must be treated for pests before distributing domestically, or exporting overseas.

Different pests need different solutions, such as chemicals, gases, or temperature treatments.

An effective alternative treatment is irradiation – technically known as phytosanitary irradiation.

After decades of research and refinements, irradiation is now used in more than 60 countries to treat food. For biosecurity purposes, irradiation is used to treat a range of pests with a single dose, and it can be used on all types of plant produce in Australia. International guidelines are in place to make sure everyone uses irradiation effectively.

Like a microwave in a kitchen, irradiation is a safe way to treat food. But, unlike a microwave, it does not heat food, so produce stays fresh. The process is fast, and easy to include in supply chains. Produce is sent to an irradiation facility, and stays in its packaging during the process, which avoids extra handling and possible damage. Irradiation passes through the packaging, treating any pests that may be on or inside the food. When the treatment is complete, the energy disappears instantly, and produce is ready to be distributed. A certificate is included to inform biosecurity inspectors that the produce is safe.

This technology is helping producers unlock more opportunities for trade in a way that's safe for both people and the environment.

Find out more about the benefits of using irradiation at agriculture.gov.au/plant-protection

Further information is available in Irradiation insights - Food irradiation for safe trade.

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[Irradiation insights - Food irradiation for safe trade \(PDF 952 KB\)](#)

[Irradiation insights - Food irradiation for safe trade \(DOCX 33 KB\)](#)

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Food irradiation for Australian producers

Title: Food irradiation for Australian producers

Australia grows many plant products – from tropical fruits and vegetables in the north, to temperate fruits and vegetables in the south. Our farmers provide fresh produce to people across the country, and overseas. But these commercial shipments can spread pests.

Trade pathways traditionally rely on chemical, gas, and temperature-based treatments. An effective alternative treatment is irradiation – technically known as phytosanitary irradiation. Unlike other treatments, irradiation is effective against a broad range of pests – including fruit fly species.

This technology is helping Australian producers unlock opportunities for trade in domestic and export markets. It also helps Australian farmers meet phytosanitary requirements without compromising the quality or sustainability of their produce. More than 60 countries treat food with irradiation and, in Australia, it can be used as a biosecurity treatment for several different types of produce.

No specialist equipment or extra work is needed at the farm – produce is treated at centralised facilities and is accessible to all businesses. The process is fast and free from heat and chemicals, making it better for produce quality and the environment.

The treatment works on full pallets of packed produce. This avoids damage from extra handling – a bonus for delicate and short-lived produce. Energy passes through the packaging, treating any pests that may be on or inside the produce. When the treatment is complete, the energy disappears instantly. A certificate is included to inform biosecurity inspectors that the produce complies with import requirements.

To start using irradiation, email MCS@aff.gov.au

Further information is available in Irradiation insights - Food irradiation for Australian producers.

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Increase your trade with food irradiation

Title: Increase your trade with food irradiation

Many countries can access fresh fruit and vegetables year-round, thanks to technology that helps grow, transport and store produce. But produce often needs to be treated for pests before it is exported. Chemicals, gases, or temperature treatments are often used. An alternative treatment is irradiation – technically known as phytosanitary irradiation. The entire treatment is fast, and free from heat and

chemicals. This maintains the quality of produce and helps get it into markets faster. It is a safe and sustainable way to provide different regions with fresh produce that they may not be able to grow in their own climates.

The treatment works on many types of plant produce and a broad range of pests, including fruit flies. Produce is conveniently treated on pallets in its final packaging before export.

The amount of irradiation is based on the requirements of the importing country. Energy passes through the package, treating any pests that may be on or inside the produce. The energy disappears instantly once the treatment is complete. A certificate is included to inform biosecurity inspectors that the consignment has been treated.

Irradiation services are highly regulated and used according to international guidelines, so countries are confident that irradiated produce they import, and export, is safe. This technology is helping unlock more opportunities for strong and safe 2-way trade around the world.

To start using irradiation, contact your national plant protection organisation.

Further information is available in Irradiation insights - Boosting opportunities for fresh produce trade among India, South-East Asia and Australia.

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[Irradiation insights - Boosting opportunities for fresh produce trade among India, South-East Asia and Australia \(DOCX 37 KB\)](#)

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Biosecurity benefits of irradiating produce

Title: Biosecurity benefits of irradiating produce

Global trade of fresh fruit and vegetables is increasing.

Different produce and pests need different treatments, such as chemicals, gases or temperatures. An effective alternative treatment is irradiation – technically known as phytosanitary irradiation.

Irradiation is used in more than 60 countries, including Australia.

It has been used for decades as a safe way to treat produce and help protect biosecurity. The process works by making insects sterile, which prevents the spread of pests.

Produce is packed at the farm and sent to an irradiation facility. Details are recorded and packages are labelled to make sure they get the right treatment – based on the produce and phytosanitary requirements of the importing country. Energy passes through the package, treating any pests that are

on or inside the produce. The energy disappears instantly once the treatment is complete. A certificate is included to inform biosecurity inspectors that the consignment has been treated.

International guidelines ensure irradiation facilities treat produce effectively and consistently, so countries are confident that irradiated produce is safe to import – regardless of where it comes from. This technology is helping producers across the world meet phytosanitary requirements and unlock more opportunities to export fresh produce.

Irradiation is fast and free from heat, chemicals and gases, making it good for delicate produce, as well as people and the environment.

To start using irradiation, contact your national plant protection organisation.

Further information is available in Irradiation insights - Biosecurity benefits of irradiating plant produce

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Include food irradiation in your produce supply chain

Title: Include food irradiation in your produce supply chain

Global trade of fresh fruit and vegetables is increasing. Before fresh produce can be exported, it often must be treated for pests to help protect the biosecurity of the importing country. Different pests and produce need different treatments, such as chemicals, gases or temperatures. A fast and convenient alternative treatment is irradiation – technically known as phytosanitary irradiation.

Irradiation is effective against a broad range of pests, including fruit flies. Treatments are conducted offsite, and no special equipment is needed at the farm. This makes it easy to include in produce supply chains. Irradiation is approved in Australia and used in more than 60 countries, helping producers across the globe access international markets.

Produce is packed at the farm and sent to an irradiation facility. Details are recorded and the process is carefully tracked, to make sure the treatment meets the phytosanitary requirements of the importing country. Irradiation energy passes through the product and packaging, treating any pests that may be on or inside the produce. When the treatment is complete, the energy instantly disappears. A certificate is included to tell biosecurity inspectors that the produce complies with import requirements.

Irradiation services are highly regulated, and international guidelines are in place to make sure everyone uses it correctly. This has helped make irradiation a reliable biosecurity treatment, trusted by producers, exporters, and importers across the world.

The whole process is fast and free from heat, chemicals and gases, making it good for produce quality, people and the environment.

To start using irradiation, contact your national plant protection organisation.

Further information is available in Irradiation insights - Streamlining produce supply chains.

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[Irradiation insights - Streamlining produce supply chains \(PDF 860 KB\)](#)

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Source: <https://www.agriculture.gov.au/agriculture-land/plant/health/international-plant-protection/irradiation-insights>